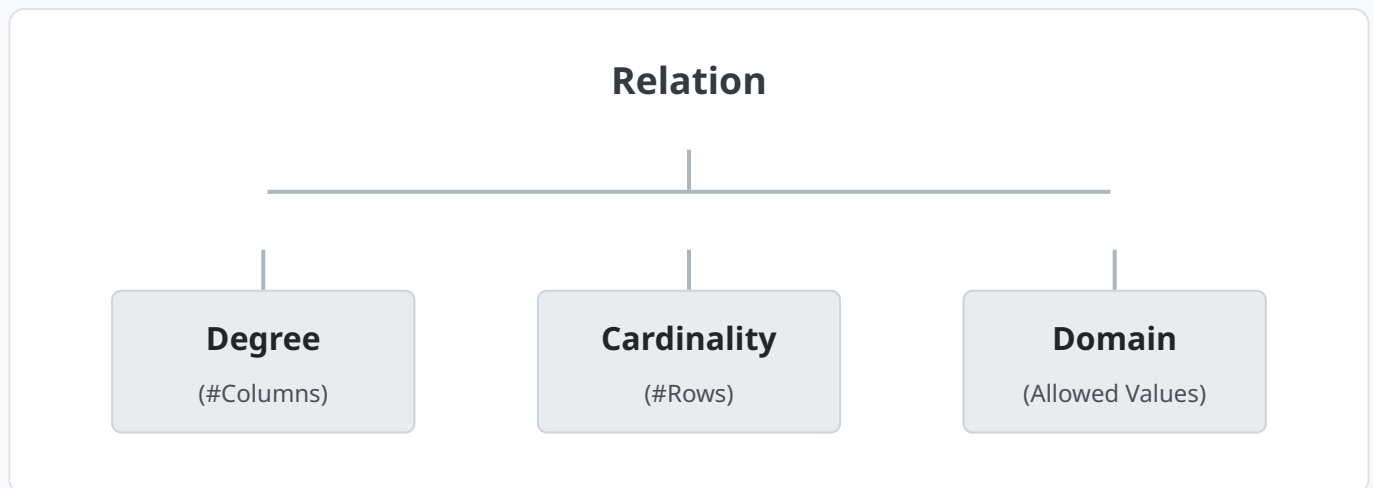


CHARACTERISTICS OF A RELATION

Every relation has three important characteristics.



Remember this forever:

Degree

→ Columns

Cardinality

→ Rows

Domain

→ Allowed values for a column .

| What is Degree?

Definition

The degree of a relation is the total number of attributes (columns) present in the relation.

Degree tells us:

- The structure of the relation
- The amount of information recorded per tuple
- Whether the schema has changed

It is part of the schema, not the data.

What is Cardinality?

Definition

The cardinality of a relation is the total number of tuples (rows) present in the relation at a given time.

THE BIGGEST INTERVIEW TRAP

There are two different meanings of cardinality in DBMS.

Meaning 1 — Relation Cardinality

This chapter.

Definition:

Number of rows.

Meaning 2 — ER Diagram Cardinality

Used in ER modeling.

Examples:

- 1 : 1
- 1 : N
- N : M

This describes how entities relate to each other, not how many rows are in a table.

What is a Domain?

This is one of the most important theoretical concepts.

Definition

A domain is the set of all permissible (allowed) values that an attribute can take.

Relation cardinality refers to the number of tuples in a relation. ER cardinality refers to the relationship multiplicity between entities, such as 1:1, 1:N, or M:N.

A data type defines the representation (e.g., INT, VARCHAR), while a domain defines the valid set of values for an attribute.